Evaluacion Learning Rate

El *learning rate* define la estabilidad y la calidad de convergencia del agente TD3, especialmente en la relación actor–critic.

Valores altos en ambos (**α_a = 1e−3, α_c = 1e−3**) generan actualizaciones agresivas, provocando inestabilidad y dificultando la convergencia de la política.

Cuando el *critic* incrementa su tasa (**α_c ≥ 5e−3**), el entrenamiento se degrada aún más, introduciendo ruido en la estimación de Q y afectando directamente al aprendizaje del actor.

Por el contrario, un *actor* con tasa baja (**α_a = 1e−5**) mejora la estabilidad, pero ralentiza excesivamente el aprendizaje, limitando la capacidad de adaptación del agente.

El mejor compromiso se obtiene con una configuración asimétrica:
α_a = 1e−4 y α_c = 1e−3, donde el critic aprende suficientemente rápido para guiar al actor, mientras que el actor actualiza de forma más estable y controlada.

El punto óptimo para TD3 se alcanza con un *actor learning rate* menor que el del *critic* (1e−4 vs 1e−3), garantizando estabilidad, buena convergencia y mejor desempeño en RMSE.

Evaluacion Learning Rate

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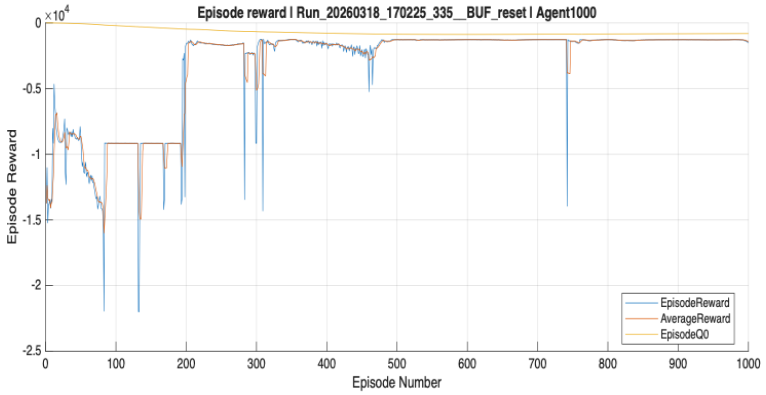
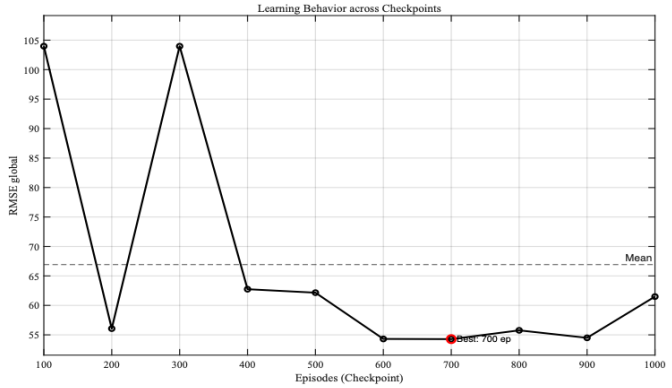
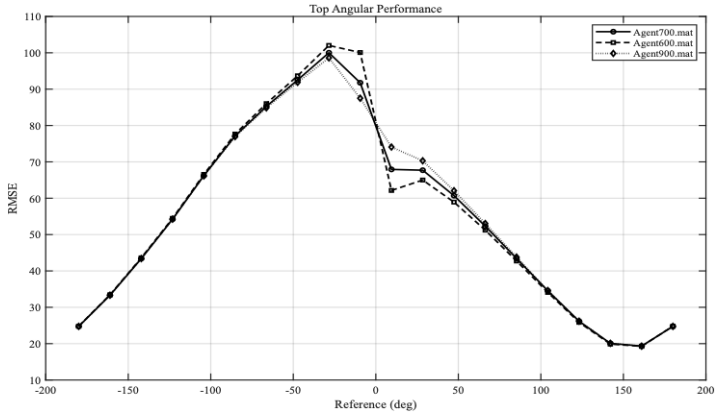
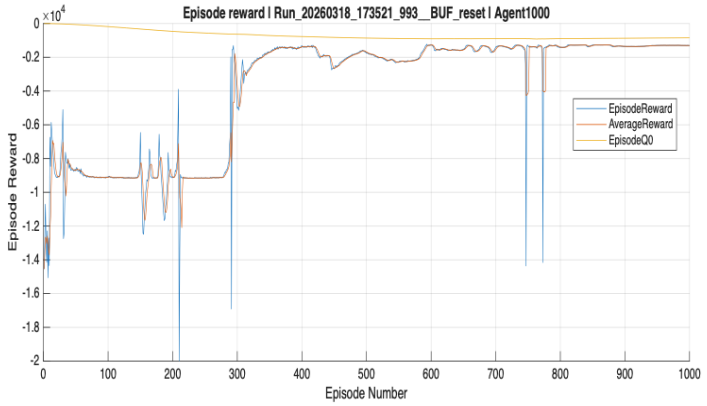
Actor LR: 1-e3,Critic LR: 1-e3

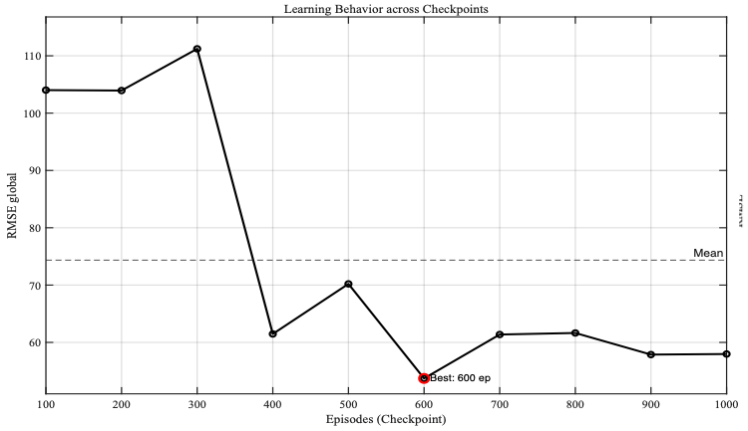


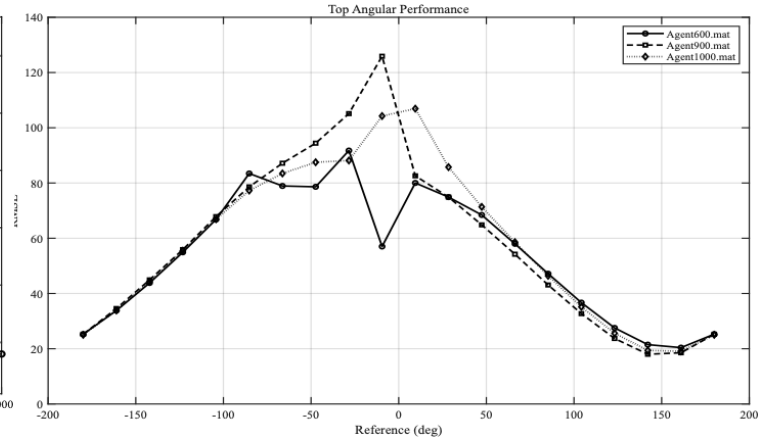




Actor LR: 1-e4,Critic LR: 1-e3







Evaluacion Learning Rate

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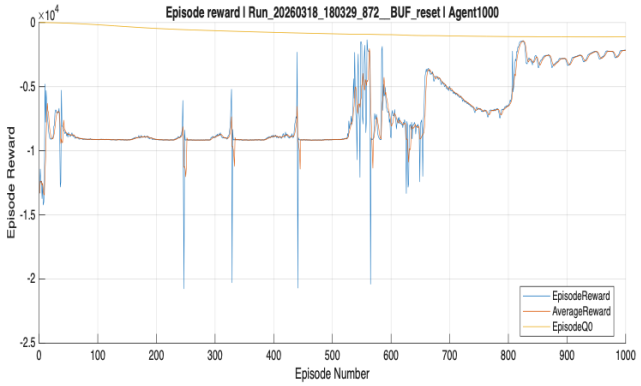
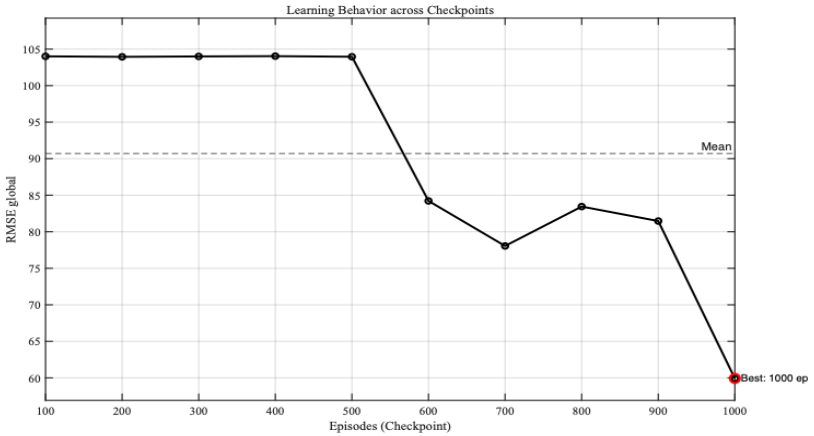
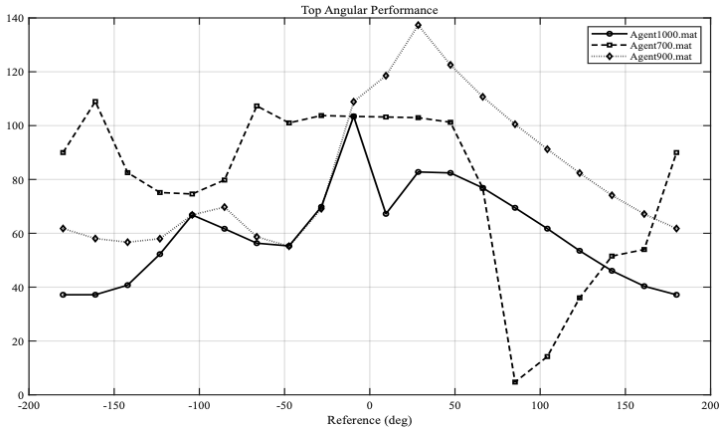
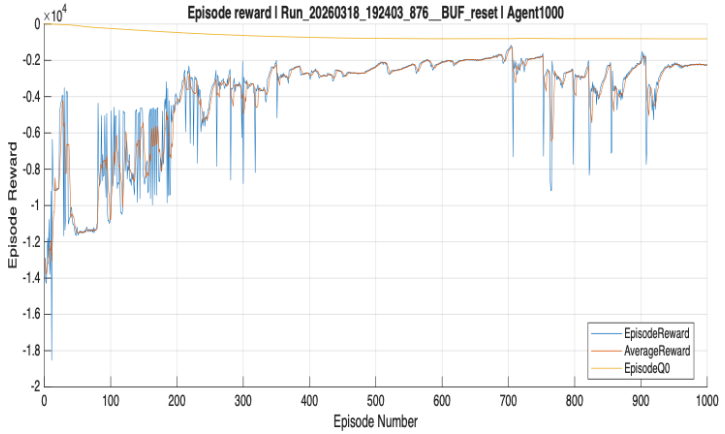
Actor LR: 1-e5,Critic LR: 1-e3

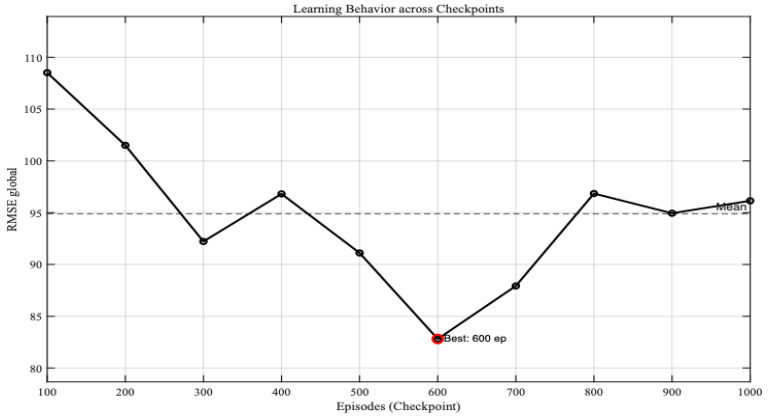


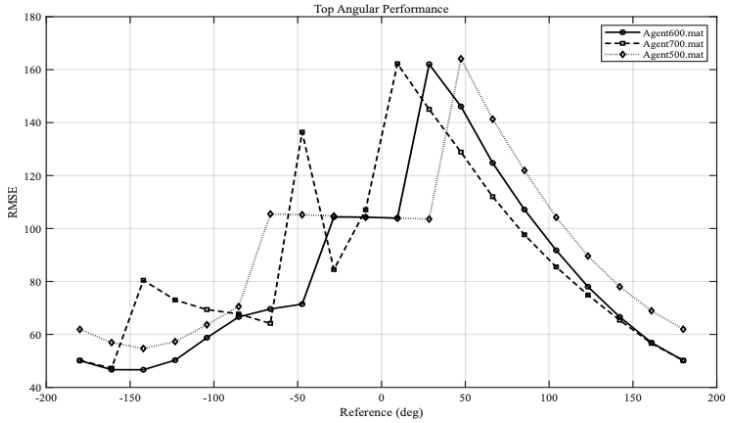




Actor LR: 1-e3,Critic LR: 5-e3







Evaluacion Learning Rate

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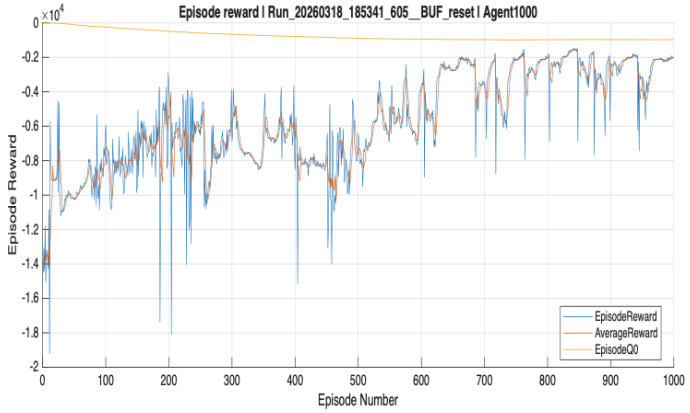
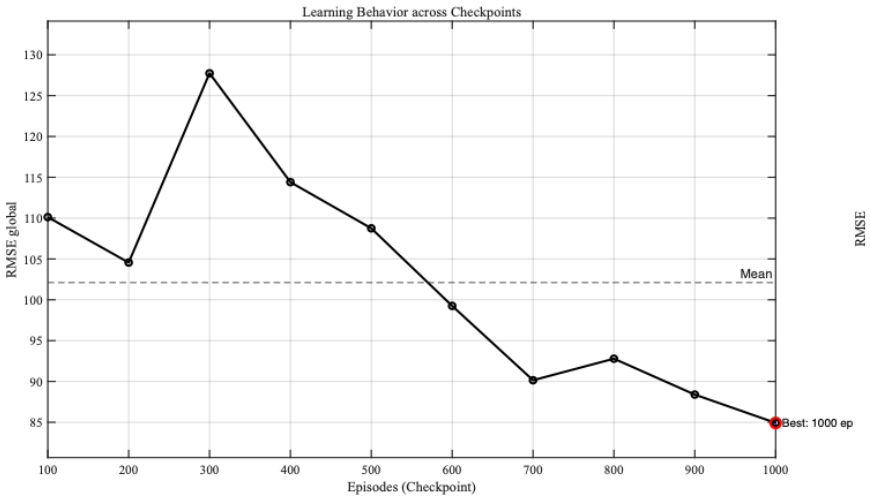
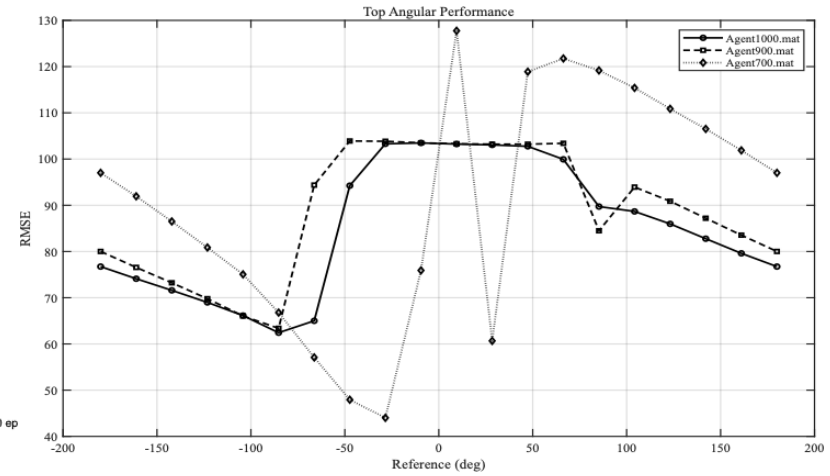
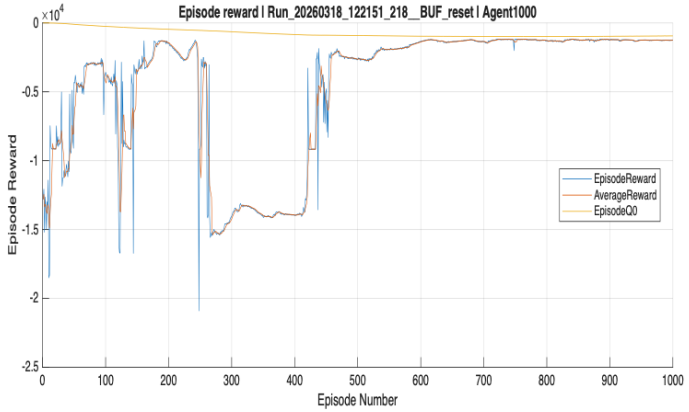
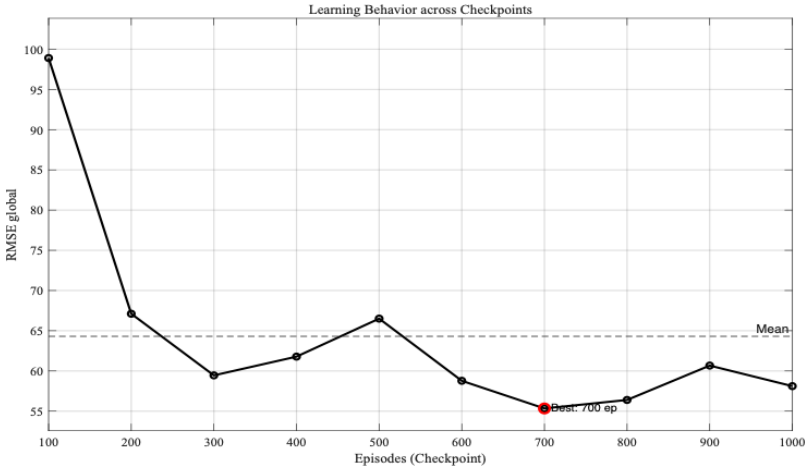
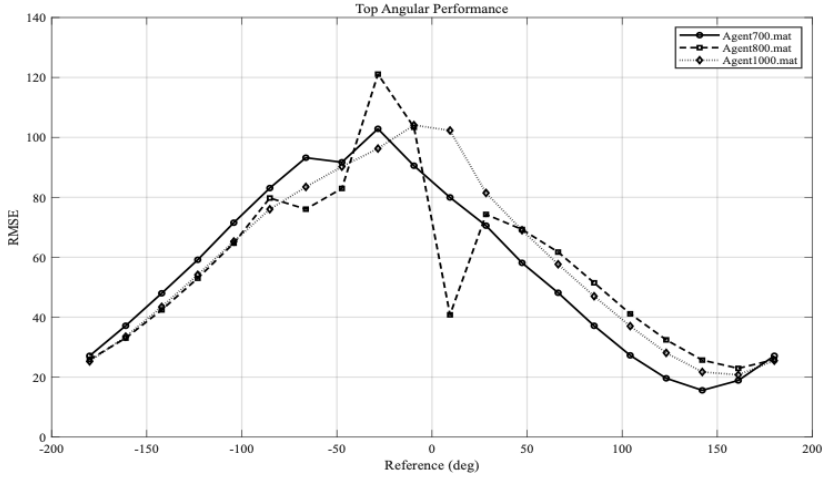
Actor LR: 1-e3,Critic LR: 6-e3







Actor LR: 1-e5,Critic LR: 7-e3